

A Spoken Dialogue Framework for Bidirectional Interaction with Robot Knowledge Graphs via Grammar-Constrained LLM Generation

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Context & Problem

Why Spoken Dialogue is Still Difficult for Cognitive Robots?

- Service robots increasingly interact through natural language.
- Dialogue should both retrieve and update robot knowledge.
- Regular LLM dialogue systems mainly generate text, not structured robot memory.
- Robot cognitive architectures require symbolic for planning and reasoning.
- Knowledge Graphs are a suitable representation, but also interpretable.



Motivation

Research Gap:

- Can a human interact with a robot's Knowledge Graph
- using spoken dialogue
- without sacrificing structure,
- consistency,
- or explainability?

Current limitations:

- Free-form LLM generation may produce unreliable structured outputs.
- Cloud-based assistants introduce latency and privacy concerns.
- Robot memory updates should remain traceable and auditable.

We propose a grammar-constrained spoken dialogue framework that safely bridges natural language and robot Knowledge Graphs.

Problem Formalization

Knowledge Graph: $\mathcal{G} = (\mathcal{N}, \mathcal{E}, \mathcal{P})$ (typed nodes, edges, properties)

KG Operations $\mathcal{O}$:

CreateNode : (n, τ) CreateEdge : (r, s, t)

RemoveNode : (n) RemoveEdge : (r, s, t)

SetProp : (x, k, v) Query : $\mathcal{G} \rightarrow \text{NL}$

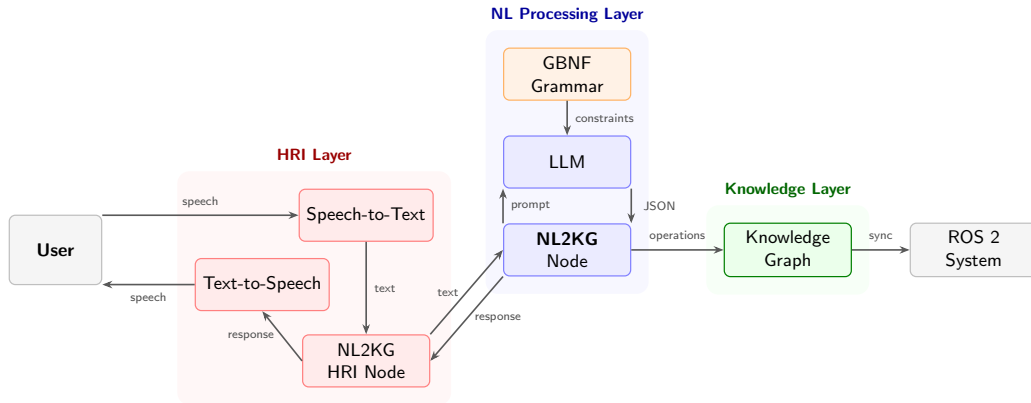
NL-to-KG Mapping: $f(u, \mathcal{G}) = (i, \mathbf{o}, r)$ (intent, operations, response)

Grammar-Constrained Generation:

$$p'(x_t \mid x_{<t}) = \begin{cases} \frac{p(x_t \mid x_{<t})}{Z} & \text{if } x_{<t} \cdot x_t \in \text{Prefix}(\mathcal{L}(G_{\text{GBNF}})) \\ 0 & \text{otherwise} \end{cases}$$

Every output is syntactically valid JSON with correct schema.

System Architecture



Evaluation

8 categories, 500 samples

Category	Samples
Node creation	75
Edge creation	75
Property setting	62
Node query	62
Edge query	62
Node removal	51
Edge removal	51
Multi-operation	62

Metrics: Intent acc., Operation F1, Exact Match, EmbSim, Latency

Evaluated models (4B class, on-robot upper bound):

- Gemma-4-4B-it
- Nemotron3-Nano-4B
- Phi-4-mini-instruct
- Qwen-3-4B
- Qwen-3.5-4B

Each under grammar-constrained and unconstrained decoding.

Grammar vs. Unconstrained Results

Model	Cond.	Intent	F1	EM	EmbSim	Lat (s)
Gemma-4-4B	+Gram.	96.6	64.3	51.2	74.6	0.96
	−Gram.	96.8	51.2	41.2	74.2	0.95
Nemotron3-4B	+Gram.	96.8	86.8	68.4	95.2	0.75
	−Gram.	97.0	67.1	63.8	93.6	0.77
Phi-4-mini	+Gram.	88.6	80.2	64.0	88.4	0.76
	−Gram.	9.4	30.4	6.2	11.4	0.52
Qwen-3-4B	+Gram.	99.0	89.9	77.0	97.8	0.78
	−Gram.	99.0	72.6	64.8	97.4	0.70
Qwen-3.5-4B	+Gram.	99.0	93.6	85.4	98.2	0.92
	−Gram.	98.8	74.8	66.6	97.6	0.92

Per-Category Analysis (Grammar-Constrained)

Category	Metrics	Gemma-4	Nemotron	Phi-4	Qwen-3	Qwen-3.5
Node creation	F1	100.0	98.2	96.4	98.2	99.3
	EM	100.0	97.3	94.7	94.7	98.7
Edge creation	F1	82.0	67.7	56.7	79.3	84.9
	EM	45.3	0.0	5.3	37.3	57.3
Property setting	F1	69.4	77.4	38.7	71.0	83.9
	EM	64.5	77.4	30.6	71.0	83.9
Node query	F1	3.2	100.0	98.4	100.0	100.0
	EM	3.2	100.0	95.2	100.0	100.0
Edge query	F1	0.0	100.0	100.0	100.0	100.0
	EM	0.0	100.0	100.0	100.0	100.0
Node removal	F1	100.0	100.0	100.0	98.0	100.0
	EM	100.0	100.0	100.0	98.0	100.0
Edge removal	F1	98.0	90.8	98.0	100.0	100.0
	EM	92.2	88.2	98.0	100.0	100.0
Multi-operation	F1	63.1	65.1	61.9	76.7	83.9
	EM	11.3	1.6	6.5	27.4	51.6



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Discussion

- **Grammar constraints:** Model-independent guarantee of structural validity.
- **4B deployment:** Practical upper bound for on-robot inference.
- **100% structural validity** with grammar; Phi-4-mini collapses without it.
- Grammar yields **F1 point, Exact Match, Embedding Similarity gains**, while latency is preserved.
- Edge creation & multi-operation show **large F1–EM gap** (partial vs. exact).
- **Qwen-3.5-4B leads overall.**

Conclusions

The framework bridges spoken language and robot KGs via grammar-constrained LLM generation.

Contributions:

- Bidirectional Speech $\leftrightarrow$ KG interface with auditable updates.
- Formal NL-to-KG mapping with grammar-constrained decoding.
- Benchmark (500 samples, 8 categories),
- Evaluation of five 4B models under constrained/unconstrained conditions.

Best results: Qwen-3.5-4B (99.0% intent, 93.6% F1, 85.4% EM, 98.2% Embedding Similarity).

Future: Human trials, multi-turn dialogue, perceptual grounding, RAG

Code: <https://github.com/mgonzs13/nl2kg>



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